

Ratios & Simplest Form — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

The simplest form of 24 : 18 is:

(a) 12 : 9 (b) 4 : 3 (c) 3 : 4 (d) 8 : 6

Q2 • MCQ**[1]**

Which ratio is equal to 3 : 2?

(a) 60 : 40 (b) 6 : 9 (c) 2 : 3 (d) 9 : 4

Q3 • FILL IN THE BLANK**[1]**

Write 600 : 900 in simplest form (type as a:b).

Q4 • MCQ**[1]**

The ratio 1 kg : 800 g in simplest form is:

(a) 1 : 800 (b) 5 : 4 (c) 4 : 5 (d) 1 : 8

Q5 • TRUE / FALSE**[1]**

The ratio 3 : 2 is the same as 2 : 3.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): The ratios 72 : 96 and 3 : 4 are proportional.

Reason (R): Both have the same simplest form, 3 : 4.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Write in simplest form: (i) 45 : 60 (ii) 35 : 14.

Q8

[2]

Are the ratios 21 : 6 and 35 : 10 proportional? Justify your answer.

Q9

[2]

A drink mixes sugar and water in the ratio 6 : 10. Write this ratio in simplest form.

Long answer (3 marks each)

Q10

[3]

Three rectangles have width $\times$ height: A = 60×40 , C = 30×20 , E = 60×60 . Which are similar to each other? Justify using ratios.

Q11

[3]

When Neelima is 3 years old her mother is 30. Write the ratio of their ages in simplest form, and again when Neelima is 12 (mother 39). Are they the same?

Q12

[3]

A wall is 2 m long and 50 cm thick. Write the ratio of length to thickness in simplest form, converting units first.

Total: 21 marks · Check your work against the answer key.